

Curtin University, Semester 2, 2026**ECOM5005 Tutorial 8***Hypothesis Testing: One-sample Tests***Learning Outcomes**

1. Identify the basic principles of hypothesis testing.
2. Explain the assumption of each hypothesis-testing procedure, how to evaluate them and the consequences if they are seriously violated.
3. Use hypothesis testing to test a mean or proportion.
4. Recognise the pitfalls involved in hypothesis testing.
5. Identify the ethical issues involved in hypothesis testing.

PART 1 – Short answer questions**Question 1**

For the hypothesis test below, where the decision is whether to reject H_0 , then a ____ error could have occurred.

$H_0: \mu \geq 90$, $H_1: \mu < 90$, $\alpha = 10\%$

Critical value = -1.28; Test value = -2.3; Decision: Reject H_0 ; Conclusion: The mean is less than 90.

A. Type I B. Type II C. None

Question 2

A survey claims that 9 out of 10 doctors recommend aspirin for their patients with headaches. To test this claim against the alternative that the actual proportion of doctors who recommend aspirin is less than 0.90, a random sample of 100 doctors results in 83 who indicate that they recommend aspirin. The value of the test statistic in this problem is approximately equal to

A. -0.07 B. -2.33 C. -1.86 D. -4.12

Question 3

A major pet store chain is considering opening a new store in an area that currently does not have any such stores. The chain will open if there is evidence that more than 5,000 of the 20,000 households in the area own pets. It conducts a telephone poll of 300 randomly selected households in the area and finds that 96 have pets.

(1) State the test of hypothesis that is of interest to the pet store chain.

A. $H_0: \pi \leq 5,000$ vs. $H_1: \pi > 5,000$ B. $H_0: \pi \leq 0.25$ vs. $H_1: \pi > 0.25$ C. $H_0: \pi \leq 0.32$ vs. $H_1: \pi > 0.32$ D. $H_0: \pi \leq 5,000$ vs. $H_1: \pi > 5,000$

(2) The value of the test statistic in this problem is approximately equal to

A. 1.94 B. 1.30 C. 2.60 D. 2.80

(3) The p-value associated with the test statistic in this problem is approximately equal to

A. 0.0100 B. 0.0026 C. 0.0051 D. 0.0013

(4) The decision on the hypothesis test using a 5% level of significance is

A. To accept H_0 in favour of H_1 B. To fail to reject H_0 in favour of H_1 C. To reject H_0 in favour of H_1 D. We cannot tell what the decision should be from the information given.

(5) The chain's conclusion from the hypothesis test using a 5% level of significance is

A. To delay opening a new store until additional evidence is collected B. Not to open a new store C. To open a new store D. We cannot tell what the decision should be from the information given.

Real-life problems**Question 4 – EV battery range and a Z test**

An electric-vehicle manufacturer advertises that the mean highway range of a new battery pack is 400 km. Long-run engineering records indicate that the population standard deviation is $\sigma = 6$ km. An independent random sample of 64 production packs has a mean range of 402.1 km.

(a) State H_0 and H_1 for testing whether the true mean range differs from 400 km.

(b) At $\alpha = 0.01$, use the critical-value approach to perform the test.

(c) Calculate the two-tail p-value and make the same decision using the p-value approach.

(d) Construct the corresponding 99% confidence interval for μ and explain its connection to the test.

(e) Explain a Type I error in this context and comment on statistical significance versus practical significance.

Question 5 – AI helpdesk improvement and a one-sample t test

A SaaS company has redesigned its AI helpdesk workflow. Historically, the mean time from a customer opening a ticket to a human handoff was 12.0 minutes. After the redesign, a random sample of 36 tickets has a mean handoff time of 11.1 minutes and a sample standard deviation of 2.4 minutes. Management wants evidence that the redesign has reduced the mean handoff time.

- (a) State H_0 and H_1 and identify whether this is a lower-tail, upper-tail or two-tail test.
- (b) At $\alpha = 0.05$, calculate the t statistic, degrees of freedom and critical value.
- (c) Calculate the p-value and make a statistical decision.
- (d) State the managerial conclusion, explain a Type I error, and comment on whether the estimated 0.9-minute reduction is practically meaningful.

Question 6 – Streaming conversion and a proportion test

A streaming platform historically converts 30% of free trials to paid subscriptions. After redesigning the onboarding process, 139 of 400 randomly selected new trials convert to paid subscriptions. The product team wants to know whether the conversion rate has increased.

- (a) State H_0 and H_1 and check the normal-approximation conditions for a one-sample proportion test.
- (b) At $\alpha = 0.05$, calculate the z statistic and the upper-tail critical value.
- (c) Calculate the p-value and state the decision at $\alpha = 0.05$.
- (d) Would the decision change at $\alpha = 0.01$? Explain.
- (e) Explain a Type I error in this context and distinguish statistical significance from business significance.

PART 2 – Data exercise

Question 7 – Consumer credit-card expenditure

This exercise uses Bill Greene's credit-card expenditure data supplied in the Excel workbook ECOM5005_Tutorial_8_Consumer_Credit_Card_Data.xlsx. The workbook contains 72 observations. AVGEXP is average monthly credit-card expenditure; OWNRENT equals 1 for a homeowner and 0 for a renter.

- (a) Open the workbook and inspect the Read_Me and "Credit_Card_Data" sheets. Do not sort or delete the raw data.
- (b) For AVGEXP, calculate n, mean, median, sample standard deviation, minimum and maximum. Create a histogram and briefly describe the shape of the distribution.
- (c) A planning benchmark states that mean monthly credit-card expenditure is \$200. At $\alpha = 0.05$, test $H_0: \mu = 200$ against $H_1: \mu \neq 200$ using a one-sample t test. Show the hypotheses, test statistic, degrees of freedom, critical value, p-value, decision and business conclusion.
- (d) Construct a 95% confidence interval for μ and explain why it should lead to the same decision as the two-tail test in part (c).
- (e) Create a frequency table and column chart for HOMEOWNER. A planning assumption is that 50% of cardholders own their home. At $\alpha = 0.05$, test $H_0: \pi \geq 0.50$ against $H_1: \pi < 0.50$. Check the normal-approximation conditions and report z, critical z, p-value, decision and conclusion.
- (f) Comment on the assumptions and limitations of the analysis, including the strong skewness in expenditure, sample representativeness, and the difference between statistical and practical significance.

Dataset source: Bill Greene's credit scoring data, distributed with statsmodels. Source documentation: <https://www.statsmodels.org/dev/datasets/generated/ccard.html>

PART 3 – Pearson MyLab

1. Please ensure that you have completed Quiz 4.

Note: If you are having issues with obtaining access to MyLab, please re-read the instructions (available in the welcome topic). If you are still unsuccessful, please send an email to the teaching team explaining the issue. It is important that we complete MyLab tasks on a weekly basis since the final e-test in this unit might be completed using MyLab.